ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q4

Question 1

(a) (i) $f_x = f / p = 16\text{ mm} / 5\ \mu\text{m} = 16\times10^{-3} / 5\times10^{-6} =$ **3200 px**.

(ii) Sensor width $W_s = 1280 \times 5\ \mu\text{m} = 6.4$ mm. By similar triangles FOV $= W_s \cdot Z / f = 6.4 \times 400 / 16 =$ **160 mm**.

Distance per pixel = FOV / 1280 = 160/1280 = **0.125 mm/px** (check: $Z/f_x = 400/3200 = 0.125$ mm/px).

(b) LSB $= V_{range}/2^n = 5\text{ V} / 2^{12} = 5/4096 = 1.2207$ mV.

Smallest resolvable force ΔF = LSB / S = 1.2207 mV / (10 mV/N) = **0.1221 N**.

(c) ZOH discretisation: $F = e^{AT}$, $G = \int_0^T e^{A\tau}B\,d\tau = A^{-1}(F - I)B$ $(A \neq 0)$.

Here $A = -2$, $B = 1$, $T = 0.1$: $F = e^{-0.2} =$ **0.8187**. $G = (F - 1)/A = (0.8187 - 1)/(-2) =$ **0.0906**.

$x_{k+1} = 0.8187\ x_k + 0.0906\ u_k$.

Question 2

(a) States $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau - b\,x_2 - mg\ell \sin x_1]$.

Equilibria ($\tau = \tau_0$, $\dot{x} = 0$): $x_2 = 0$ and $mg\ell \sin\theta_e = \tau_0$. Sence $|\sin\theta| \le 1$, an equilibrium exists iff $|\tau_0| \le mg\ell = 2.4525$ N·m.

(b) $\delta\dot{x} = A\,\delta x + B\,\delta\tau$, $A = \partial f/\partial x$, $B = \partial f/\partial\tau$ at $\theta_e = 60°$, $x_{2e} = 0$.

$A = [0,\ 1;\ -(mg\ell/J)\cos\theta_e,\ -b/J] = [0,\ 1;\ -122.6\times0.5,\ -2.5] =$ **[0, 1; −61.31, −2.5]**.

$B = [0;\ 1/J] =$ **[0; 50]**.

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$, so the PD torque must balance gravity: $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$.

If $\theta_{ss} = \theta_d = 90°$ the left side is 0 but the right side is $mg\ell \sin 90° = 2.4525 \neq 0$: contradiction. So the joint cannot rest at θ_d; the gravity torque needs a non-zero error to produce the holding torque.

Solve $50(\pi/2 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$ numerically: $\theta_{ss} \approx 1.5218$ rad = 87.19°. Error ≈ 90° − 87.19° = **2.81°** (joint sags below θ_d).

(d) With compensation, at rest: $K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss} = mg\ell \sin\theta_{ss} \Rightarrow \theta_{ss} = \theta_d$: **steady-state error = 0**.

Closed loop: $J\ddot{\theta} + (b + K_d)\dot{\theta} + K_p(\theta - \theta_d) = 0$; gravity cancels. Without compensaton the linearised stiffness is $K_p + mg\ell \cos\theta_d$, and at $\theta_d = 90° \cos\theta_d = 0$, so the stiffness is K_p either way. **Poles unchanged** (specific to 90°; at other angles they would shift).

Question 4

(a) $\ddot{\theta} + a\dot{\theta} + c\theta = b(N_{ff}\,r - k_1\theta - k_2\dot{\theta}) \Rightarrow \ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$.

With $a = 2$, $b = 1$, $c = 6$, $k_1 = k_2 = 2$: $s^2 + 4s + 8 = 0 \Rightarrow s = -2 \pm 2j$ (stable).

DC gain $= bN_{ff}/(c + bk_1) = N_{ff}/8 = 1 \Rightarrow$ **$N_{ff} = 8$**.

(b) The acuator is 25% weaker, so the output is also 25% lower: $\theta_{ss} = 0.75$, and the tracking error = **25%**.

(c) $\ddot{\theta} + a\dot{\theta} + c\theta = b(-k_1\theta - k_2\dot{\theta} + K_I\xi)$, $\dot{\xi} = r - \theta$. Take $r = 0$ and differentiate once, using $\dot{\xi} = -\theta$: $d^3\theta/dt^3 + (a + bk_2)\ddot{\theta} + (c + bk_1)\dot{\theta} + bK_I\theta = 0$.

Char. poly: $\mathbf{s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I}$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

Match: $2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$; $6 + k_1 = 24 \Rightarrow \mathbf{k_1 = 18}$; $K_I = 32$ ($b = 1$).

(d) For a constant r, in steady state all derivatives vanish: $\dot{\xi} = r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$, independent of b and c; the closed loop only needs to be stable so that a steady state exists.

Failure mode: integrator windup. When the actuator saturates the error persists, ξ keeps growing and causes a large overshoot afterwards. Remedy: anti-windup (clamp the integrator, conditional integration or back-calculation).